

Hugo Garrido-Lestache Belinchon

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SUMMARY

- Interested in AI, Reinforcement Learning and cooperative multi-agent systems and **1st** in the NVIDIA ROSIE Supercomputer Challenge. Published Author attending ICMLA 2025.
- Support me on YouTube at [@HugoDoesAi](#) and come checkout my [WEBSITE](#).

EDUCATION

M.S. Machine Learning	Milwaukee School of Engineering	Jan 2025 - Aug2026
B.S. Computer Science (GPA: 3.95)	Milwaukee School of Engineering	Sep 2022 - Dec 2025
Game Design & Game Theory Course	University of Lucca	June 2023

RESEARCH EXPERIENCE

Undergraduate Research Assistant | Published at ICMLA 2025 Jan 2025 – Present

Advisor: [Jeremy Kedziora, Ph.D.](#)

[Paper](#) | [Blog](#) | [Talk](#) | [Code](#)

- Developed a TAAC, a novel Multi-Agent Reinforcement Learning algorithm to promote collaboration.
- Utilized self-play, curriculum learning and a novel algorithm to achieve state-of-the-art performance in a soccer environment and tower building environment.
- Came **1st** in the MSOE NVIDIA ROSIE Super-Computer Competition winning over \$9,000 in prizes, presenting to VP of Software and VP of Medical Imaging.

INDEPENDENT RESEARCH PROJECTS

Research on Developing Education AI Tools

[Paper](#)

Collaborators: [Ethan Jeffers](#) & [Ethan Wrasman](#)

- Developed various AI tools to enhance learning in early programming Education through LLMs.
- Statistically tested these tools to ensure their effectiveness and prevent potential misuse.

Research on Silent Video Sound Generation | Published at MICS 2024 (Oral)

[Paper](#)

Collaborators: [Adam Haile](#) & [Helina Mulugeta](#)

- Utilized VQ-VAE to be able to embed and map silent videos to sound.

WORK EXPERIENCE

Data Science Intern | Company: Direct Supply

November 2023 – Present

- Ideated and Tested AI tools to solve problems. Heavily utilized RAGs, LLMs and AWS to build Full-Stack applications and prototypes, below are a select example:
- **Reverse Image Search:** Prototyped a system to search for products by image, allowing for faster product retrieval and matching.
- **AI Product Linking:** Improved AI systems to match products together based on semantic similarity using embeddings and a graph-network (Trade Secret).
- **AI Product Tagging:** Created a System to add tags/attributes to products automatically based on product data and instructions. This data can be used to group and filter products and feed future AI Models.
- **AI Product Web Scraper:** Created an AI tool to search, parse and combine unstructured data on the internet to augment the company's quality and quantity of product data.

PROJECTS

Wordle Solver AI

[Blog](#) | [Code](#)

- Created an AI solver using Information Theory which achieves an average of **3.8 guesses** per game.
- Made optimizations to allow for faster computer on mobile device, by pruning potential words based on letter frequency.
- Created a test suite to test performance to promote faster iteration during development.

Snake Game Reinforcement Learning AI

[Blog](#) | [Code](#)

- Created a Q-learning AI agent to learn to play the game achieving a **high score of 140**.
- Improved results by simplifying state & action space and tuning rewards and hyperparameters

Large Language Model - HugoGPT

[Blog](#) | [Code](#)

- Implemented an LLM from scratch using pytorch and trained it on a supercomputer.
- Results showcased correct grammar and punctuation, but failed to have coherent sentences due to lack of data, hardware and training time.

Emotion Recognition from Facial Images

[Blog](#)

- Built a CNN for facial emotion recognition (e.g., happy, surprise, neutral) from images.
- Iterated on data gathering, preprocessing, model training, and evaluation to improve accuracy.

MENTORING & ACTIVITIES

AI Club Mentor | MSOE AI Club

Oct 2024 – Present

- Mentoring Undergraduates on AI concepts and algorithms.
- 2025 members ended up learning crucial techniques in developing ML models and created a CNN.
- 2026 members developing various projects such as agentic video segmentation, stock market prediction, and coloring grayscale images.

Programming Tutor | Milwaukee School of Engineering

Sep 2023 – Dec 2023

- Working weekly with around 20 different students to explain topics and concepts in Programming.

Adult Scout Leader | UK Scouts | Sep 2020 - Jun 2022

Tau Beta Pi Member | Tau Beta Pi | Apr 2025 - Present

Scouts | UK Scouts | 2014 - 2021

Climber | Climbing Club | Sep 2019 - Present

AWARDS

1st in NVIDIA ROSIE Supercomputer Challenge | NVIDIA | [LINK](#)

Invitation to AI Engineer Code Summit | AI Engineer | [LINK](#)

6x Dean's List with High Honors Award | Milwaukee School of Engineering | [LINK](#)

5th in MICS Programming Competition | Midwest Instruction and Computing Symposium

Silver Duke of Edinburg Award | Duke of Edinburg

Fundamentals of Deep Learning | NVIDIA | [LINK](#)

Mr. MSOE 2023 | Society of Women Engineers

A* - Maths & A* - Computer Science & A - Physics | English A-Levels

INTERESTS

Guitar | Climbing | Puzzles | Yo-yo | Cooking | Unicycle | Gaming | Youtube